

# Session 5: Pandas

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## 1 Session 5 - pandas

| Stations_id | Stationsname            | geogr. Breite | geogr. Laenge | Stationshoehe | Bundesland     |
|-------------|-------------------------|---------------|---------------|---------------|----------------|
| 1           | Aach                    | 47.841299     | 8.849299      | 478.0         | Baden-Württem  |
| 2           | Aachen (Kläranlage)     | 50.806581     | 6.099638      | 138.0         | Nordrhein-West |
| 3           | Aachen                  | 50.782659     | 6.094087      | 202.0         | Nordrhein-West |
| 4           | Aachen-Brand            | 50.768313     | 6.120705      | 243.0         | Nordrhein-West |
| 6           | Aalen-Unterrombach      | 48.836072     | 10.059764     | 455.0         | Baden-Württem  |
| ...         | ...                     | ...           | ...           | ...           | ...            |
| 19697       | Wassertrüdingen/Wörnitz | 49.046341     | 10.602684     | 435.0         | Bayern         |
| 19698       | Werda/Vogtland          | 50.448020     | 12.305168     | 595.0         | Sachsen        |
| 19701       | Wertingen/Zusam         | 48.557500     | 10.674167     | 430.0         | Bayern         |
| 19706       | Oberndorf am Neckar     | 48.285077     | 8.576395      | 516.0         | Baden-Württem  |
| 19781       | Ingolstadt              | 48.742909     | 11.423324     | 367.0         | Bayern         |

```
Index(['Stationsname', 'geogr. Breite', 'geogr. Laenge', 'Stationshoehe',  
      'Bundesland', 'Unnamed: 6'],  
      dtype='str')
```

```
<class 'pandas.DataFrame'>
```

```
Index: 5740 entries, 1 to 19781
```

```
Data columns (total 6 columns):
```

| #   | Column        | Non-Null Count | Dtype   |
|-----|---------------|----------------|---------|
| --- | -----         | -----          | -----   |
| 0   | Stationsname  | 5740 non-null  | str     |
| 1   | geogr. Breite | 5740 non-null  | float64 |
| 2   | geogr. Laenge | 5740 non-null  | float64 |
| 3   | Stationshoehe | 5740 non-null  | float64 |
| 4   | Bundesland    | 5740 non-null  | str     |
| 5   | Unnamed: 6    | 0 non-null     | float64 |

```
dtypes: float64(4), str(2)
```

```
memory usage: 313.9 KB
```

|             | Stationsname        | geogr. Breite | geogr. Laenge | Stationshoehe | Bundesland          |
|-------------|---------------------|---------------|---------------|---------------|---------------------|
| Stations_id |                     |               |               |               |                     |
| 1           | Aach                | 47.841299     | 8.849299      | 478.0         | Baden-Württemberg   |
| 2           | Aachen (Kläranlage) | 50.806581     | 6.099638      | 138.0         | Nordrhein-Westfalen |
| 3           | Aachen              | 50.782659     | 6.094087      | 202.0         | Nordrhein-Westfalen |
| 4           | Aachen-Brand        | 50.768313     | 6.120705      | 243.0         | Nordrhein-Westfalen |
| 6           | Aalen-Unterrombach  | 48.836072     | 10.059764     | 455.0         | Baden-Württemberg   |

|             | Stationsname            | geogr. Breite | geogr. Laenge | Stationshoehe | Bundesland    |
|-------------|-------------------------|---------------|---------------|---------------|---------------|
| Stations_id |                         |               |               |               |               |
| 19697       | Wassertrüdingen/Wörnitz | 49.046341     | 10.602684     | 435.0         | Bayern        |
| 19698       | Werda/Vogtland          | 50.448020     | 12.305168     | 595.0         | Sachsen       |
| 19701       | Wertingen/Zusam         | 48.557500     | 10.674167     | 430.0         | Bayern        |
| 19706       | Oberndorf am Neckar     | 48.285077     | 8.576395      | 516.0         | Baden-Württem |
| 19781       | Ingolstadt              | 48.742909     | 11.423324     | 367.0         | Bayern        |

delete last column: (unnamed) axis=1 columns, axis=0 rows

```
Index(['Stationsname', 'geogr. Breite', 'geogr. Laenge', 'Stationshoehe',
      'Bundesland'],
      dtype='str')
```

|             | Stationsname        | geogr. Breite | geogr. Laenge | Stationshoehe | Bundesland          |
|-------------|---------------------|---------------|---------------|---------------|---------------------|
| Stations_id |                     |               |               |               |                     |
| 1           | Aach                | 47.841299     | 8.849299      | 478.0         | Baden-Württemberg   |
| 2           | Aachen (Kläranlage) | 50.806581     | 6.099638      | 138.0         | Nordrhein-Westfalen |
| 3           | Aachen              | 50.782659     | 6.094087      | 202.0         | Nordrhein-Westfalen |
| 4           | Aachen-Brand        | 50.768313     | 6.120705      | 243.0         | Nordrhein-Westfalen |
| 6           | Aalen-Unterrombach  | 48.836072     | 10.059764     | 455.0         | Baden-Württemberg   |

index location, row then column

'Aachen'

slicing by index, upper band is not included

|             | Stationsname        | geogr. Breite | geogr. Laenge | Stationshoehe | Bundesland          |
|-------------|---------------------|---------------|---------------|---------------|---------------------|
| Stations_id |                     |               |               |               |                     |
| 1           | Aach                | 47.841299     | 8.849299      | 478.0         | Baden-Württemberg   |
| 2           | Aachen (Kläranlage) | 50.806581     | 6.099638      | 138.0         | Nordrhein-Westfalen |
| 3           | Aachen              | 50.782659     | 6.094087      | 202.0         | Nordrhein-Westfalen |
| 4           | Aachen-Brand        | 50.768313     | 6.120705      | 243.0         | Nordrhein-Westfalen |
| 6           | Aalen-Unterrombach  | 48.836072     | 10.059764     | 455.0         | Baden-Württemberg   |

|             | geogr. Breite | geogr. Laenge |
|-------------|---------------|---------------|
| Stations_id |               |               |
| 1           | 47.841299     | 8.849299      |
| 2           | 50.806581     | 6.099638      |
| 3           | 50.782659     | 6.094087      |
| 4           | 50.768313     | 6.120705      |
| 6           | 48.836072     | 10.059764     |
| ...         | ...           | ...           |
| 19697       | 49.046341     | 10.602684     |
| 19698       | 50.448020     | 12.305168     |
| 19701       | 48.557500     | 10.674167     |
| 19706       | 48.285077     | 8.576395      |
| 19781       | 48.742909     | 11.423324     |

slicing by label, upper band is included

```
Stations_id
3          Aachen
4      Aachen-Brand
6      Aalen-Unterrombach
Name: Stationsname, dtype: str
```

|             | geogr. Breite | geogr. Laenge |
|-------------|---------------|---------------|
| Stations_id |               |               |
| 1           | 47.841299     | 8.849299      |
| 2           | 50.806581     | 6.099638      |
| 3           | 50.782659     | 6.094087      |
| 4           | 50.768313     | 6.120705      |
| 6           | 48.836072     | 10.059764     |
| ...         | ...           | ...           |
| 19697       | 49.046341     | 10.602684     |
| 19698       | 50.448020     | 12.305168     |
| 19701       | 48.557500     | 10.674167     |
| 19706       | 48.285077     | 8.576395      |
| 19781       | 48.742909     | 11.423324     |

```
Stations_id
1      Baden-Württemberg
2      Nordrhein-Westfalen
3      Nordrhein-Westfalen
4      Nordrhein-Westfalen
6      Baden-Württemberg
...
19697      Bayern
19698      Sachsen
19701      Bayern
19706      Baden-Württemberg
19781      Bayern
```

Name: Bundesland, Length: 5740, dtype: str

Stations\_id

1           Baden-Württemberg  
2           Nordrhein-Westfalen  
3           Nordrhein-Westfalen  
4           Nordrhein-Westfalen  
6           Baden-Württemberg

...

19697           Bayern  
19698           Sachsen  
19701           Bayern  
19706           Baden-Württemberg  
19781           Bayern

Name: Bundesland, Length: 5740, dtype: str

|                        | Stationsname | geogr. Breite | geogr. Laenge | Stationshoehe |
|------------------------|--------------|---------------|---------------|---------------|
| Bundesland             |              |               |               |               |
| Bremen                 | 15           | 15            | 15            | 15            |
| Hamburg                | 19           | 19            | 19            | 19            |
| Berlin                 | 29           | 29            | 29            | 29            |
| Saarland               | 54           | 54            | 54            | 54            |
| Schleswig-Holstein     | 223          | 223           | 223           | 223           |
| Mecklenburg-Vorpommern | 237          | 237           | 237           | 237           |
| Rheinland-Pfalz        | 264          | 264           | 264           | 264           |
| Brandenburg            | 302          | 302           | 302           | 302           |
| Sachsen                | 356          | 356           | 356           | 356           |
| Thüringen              | 391          | 391           | 391           | 391           |
| Sachsen-Anhalt         | 411          | 411           | 411           | 411           |
| Hessen                 | 436          | 436           | 436           | 436           |
| Nordrhein-Westfalen    | 436          | 436           | 436           | 436           |
| Niedersachsen          | 611          | 611           | 611           | 611           |
| Baden-Württemberg      | 619          | 619           | 619           | 619           |
| Bayern                 | 1337         | 1337          | 1337          | 1337          |

'Bayern'

Stationsname   Albaching  
geogr. Breite   48.109386  
geogr. Laenge   12.104938  
Stationshoehe   513.0  
Bundesland       Bayern  
Name: 65, dtype: object

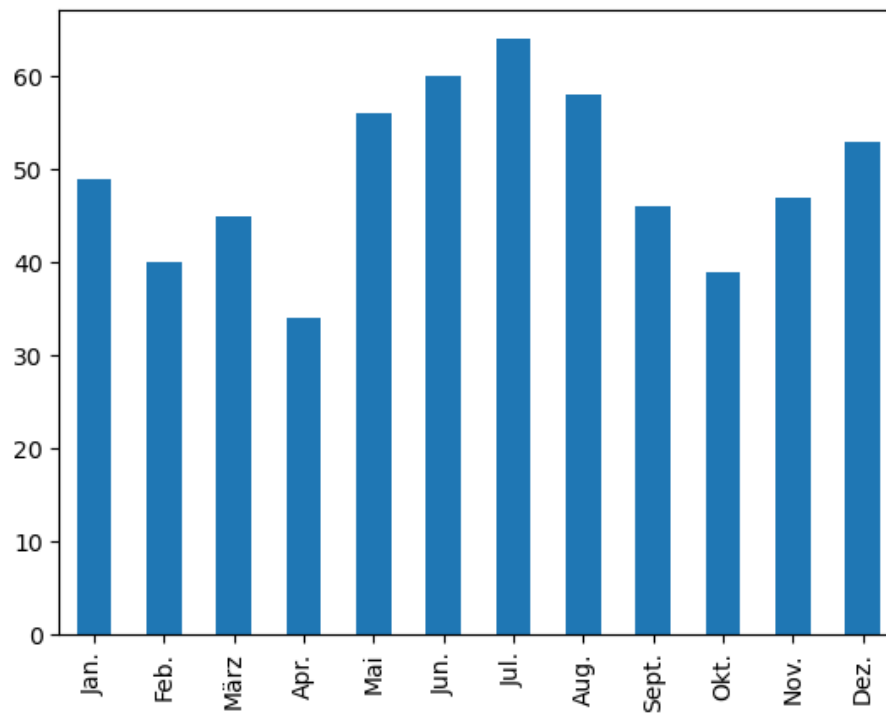
Task for this session:

Stationsname   Berlin-Malchow  
geogr. Breite   52.575287  
geogr. Laenge   13.483243  
Stationshoehe   53.0  
Bundesland       Berlin

Name: 419, dtype: object

Assignment: load the 30 year mean precipitation data data/Niederschlag 1981-2010.txt • only keep the columns with the monthly means

Monthly climate data:



Assignment: load the hourly data

|   | STATIONS_ID | MESS_DATUM | QUALITAETS_NIVEAU | STRUKTUR_VERSION | NIEDERSCHLAG |
|---|-------------|------------|-------------------|------------------|--------------|
| 0 | 399         | 2015082001 | 3                 | 0                | 0.0          |
| 1 | 399         | 2015082002 | 3                 | 0                | 0.0          |
| 2 | 399         | 2015082003 | 3                 | 0                | 0.0          |
| 3 | 399         | 2015082004 | 3                 | 0                | 0.0          |
| 4 | 399         | 2015082005 | 3                 | 0                | 0.0          |

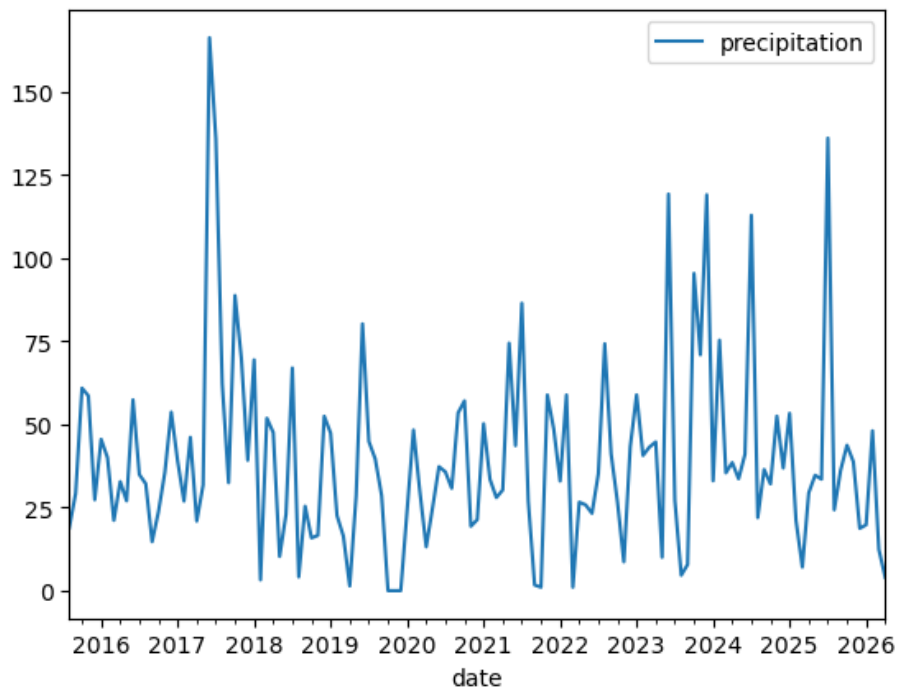
| precipitation       |     |
|---------------------|-----|
| date                |     |
| 2015-08-20 01:00:00 | 0.0 |
| 2015-08-20 02:00:00 | 0.0 |
| 2015-08-20 03:00:00 | 0.0 |
| 2015-08-20 04:00:00 | 0.0 |
| 2015-08-20 05:00:00 | 0.0 |
| ...                 | ... |
| 2026-04-12 20:00:00 | 0.0 |

|                     | precipitation |
|---------------------|---------------|
| date                |               |
| 2026-04-12 21:00:00 | 0.0           |
| 2026-04-12 22:00:00 | 0.0           |
| 2026-04-12 23:00:00 | 0.0           |
| 2026-04-13 00:00:00 | 0.0           |

|       | precipitation |
|-------|---------------|
| count | 129.000000    |
| mean  | 39.528372     |
| std   | 28.812164     |
| min   | 0.000000      |
| 25%   | 22.910000     |
| 50%   | 33.640000     |
| 75%   | 48.600000     |
| max   | 166.300000    |

|      | precipitation |           |           |       |         |        |          |        |
|------|---------------|-----------|-----------|-------|---------|--------|----------|--------|
|      | count         | mean      | std       | min   | 25%     | 50%    | 75%      | max    |
| date |               |           |           |       |         |        |          |        |
| 1    | 11.0          | 43.080909 | 14.963813 | 19.80 | 33.0000 | 45.530 | 51.7750  | 69.42  |
| 2    | 11.0          | 38.052727 | 19.801915 | 3.24  | 24.7600 | 39.990 | 48.1950  | 75.33  |
| 3    | 11.0          | 26.577273 | 16.599816 | 1.02  | 14.3900 | 27.990 | 39.2900  | 51.87  |
| 4    | 11.0          | 26.310909 | 15.251537 | 1.41  | 17.0500 | 29.420 | 35.6550  | 47.56  |
| 5    | 10.0          | 30.115000 | 17.810316 | 10.03 | 25.2225 | 27.720 | 33.1750  | 74.44  |
| 6    | 10.0          | 62.480000 | 46.832982 | 22.91 | 34.4625 | 42.310 | 74.5300  | 166.30 |
| 7    | 10.0          | 71.625000 | 43.327513 | 27.16 | 35.3325 | 55.925 | 106.2875 | 136.10 |
| 8    | 11.0          | 30.906364 | 21.515000 | 4.12  | 20.4300 | 27.030 | 35.8800  | 74.27  |
| 9    | 11.0          | 27.925455 | 15.035998 | 1.70  | 20.0650 | 29.190 | 36.3100  | 53.48  |
| 10   | 11.0          | 40.480909 | 32.242088 | 0.00  | 19.8100 | 32.090 | 58.9950  | 95.41  |
| 11   | 11.0          | 39.063636 | 25.046306 | 0.00  | 17.9900 | 38.760 | 58.7200  | 70.93  |
| 12   | 11.0          | 41.870909 | 30.349011 | 0.00  | 24.3400 | 39.200 | 50.5300  | 119.06 |

Monthly data from Alexanderplatz: (to compare with 30 year mean precipitation data)

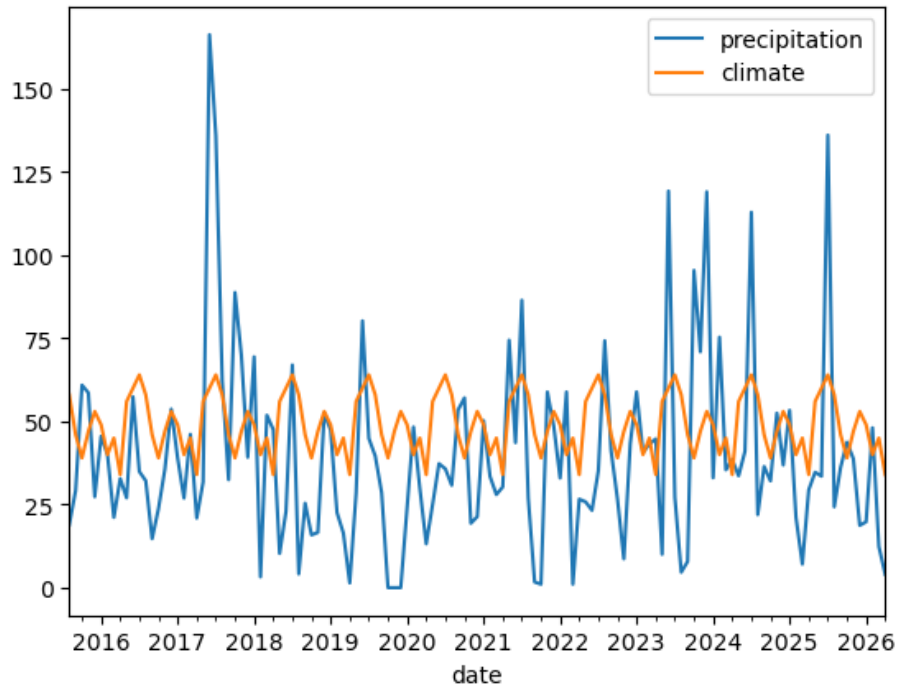


Combining with previous plot

| precipitation |      |
|---------------|------|
| 2015-01-31    | 49.0 |
| 2015-02-28    | 40.0 |
| 2015-03-31    | 45.0 |
| 2015-04-30    | 34.0 |
| 2015-05-31    | 56.0 |
| ...           | ...  |
| 2026-08-31    | 58.0 |
| 2026-09-30    | 46.0 |
| 2026-10-31    | 39.0 |
| 2026-11-30    | 47.0 |
| 2026-12-31    | 53.0 |

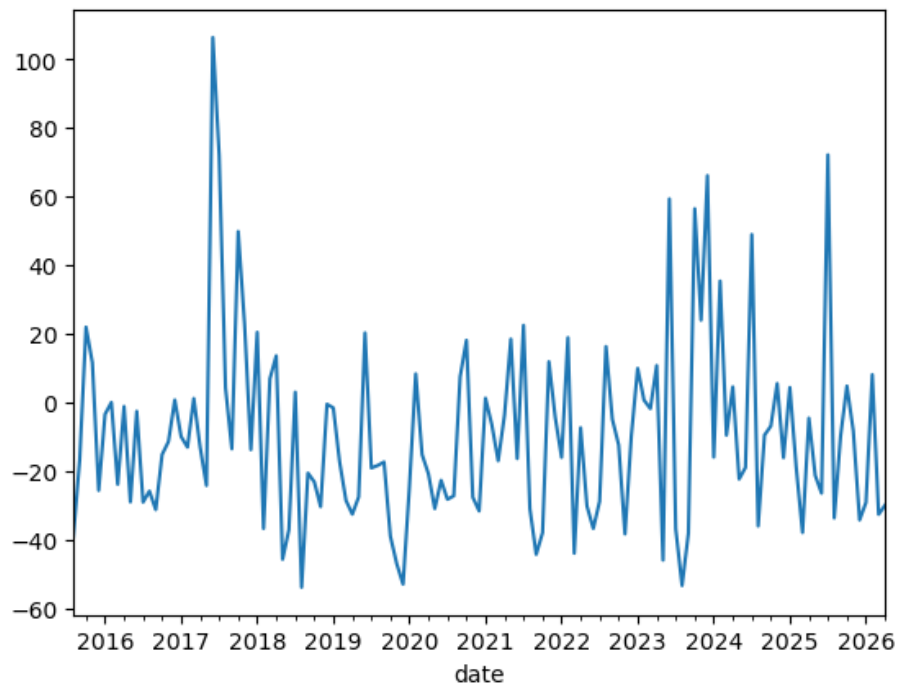
|            | precipitation | climate |
|------------|---------------|---------|
| date       |               |         |
| 2015-08-31 | 18.90         | 58.0    |
| 2015-09-30 | 29.19         | 46.0    |
| 2015-10-31 | 60.91         | 39.0    |
| 2015-11-30 | 58.59         | 47.0    |
| 2015-12-31 | 27.34         | 53.0    |
| ...        | ...           | ...     |
| 2025-12-31 | 18.73         | 53.0    |
| 2026-01-31 | 19.80         | 49.0    |

|            | precipitation | climate |
|------------|---------------|---------|
| date       |               |         |
| 2026-02-28 | 48.08         | 40.0    |
| 2026-03-31 | 12.36         | 45.0    |
| 2026-04-30 | 4.00          | 34.0    |



It is getting drier than before





Homework Session 5: I don't know why this is adding anomaly to the first plot.  
creating multiple plots w/ pyplot.subplot

